

PRESERVING FISCAL STABILITY

IN THE AGE OF TRANSFORMATIVE AI

Korinek & Lockwood | Essay #5

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Microeconomics of Artificial Intelligence | April 20, 2025

**As value creation shifts from labor to capital,
how must the U.S. tax system evolve?**

WHY IT MATTERS

- Military & defense funding
- Infrastructure investment
- Social safety net programs
- America's global competitiveness

GANS CONNECTION

AI improves prediction quality: Gans Ch. 3 frames AI as a prediction machine that makes decisions cheaper and better. This essay asks: How can the government adapt to capture that value?

THE FISCAL MATH: LABOR → CAPITAL

10–15¢

lost per \$1 shifted
from labor to capital

Effective Tax Rate Gap

LABOR INCOME: ~30% effective rate

CAPITAL INCOME: ~15–20% rate

→ *Half the revenue*

CONCRETE EXAMPLE: THE RADIOLOGIST

Human Radiologist earns \$500K/yr → generates ~\$200K in federal + state tax revenue

AI replaces them → diagnostic value is preserved (or even improves)

But if the AI company reinvests profits rather than paying wages or dividends:

Tax revenue drops to a fraction of the original amount

A PROPOSED TWO-PHASE ECONOMIC TRANSITION

PHASE 1

Twilight of Labor

AI handles tasks across the economy — customer service, legal research, medical diagnostics

Labor income tax revenue declines as a share of GDP

Key tension:

Economy grows, but government's ability to fund public goods shrinks

Gans Ch. 4: AI is a substitute for rules-based decisions. Firms moving from rules to decisions reduce predictable employment patterns.



PHASE 2

AI-Dominated Economy

AI companies reinvest most profits to expand capabilities rather than pay wages or dividends

Both labor and consumption tax bases erode

The "forest" metaphor:

Economy keeps growing but value is hard to "harvest" through traditional tax channels

Labor share fell from 68% (2000) to 58% (2025), costing hundreds of billions in annual federal revenue. AI threatens to accelerate this (i.e. increase layoffs).

Shift from taxing what people earn → taxing what people spend

VALUE-ADDED TAX (VAT)

Used by virtually every high-income country except the U.S.

Self-enforcing: businesses deduct VAT paid to suppliers, creating a paper trail that makes evasion difficult

Harder to shift offshore than income taxes which protects American competitiveness

THE KEY PRINCIPLE

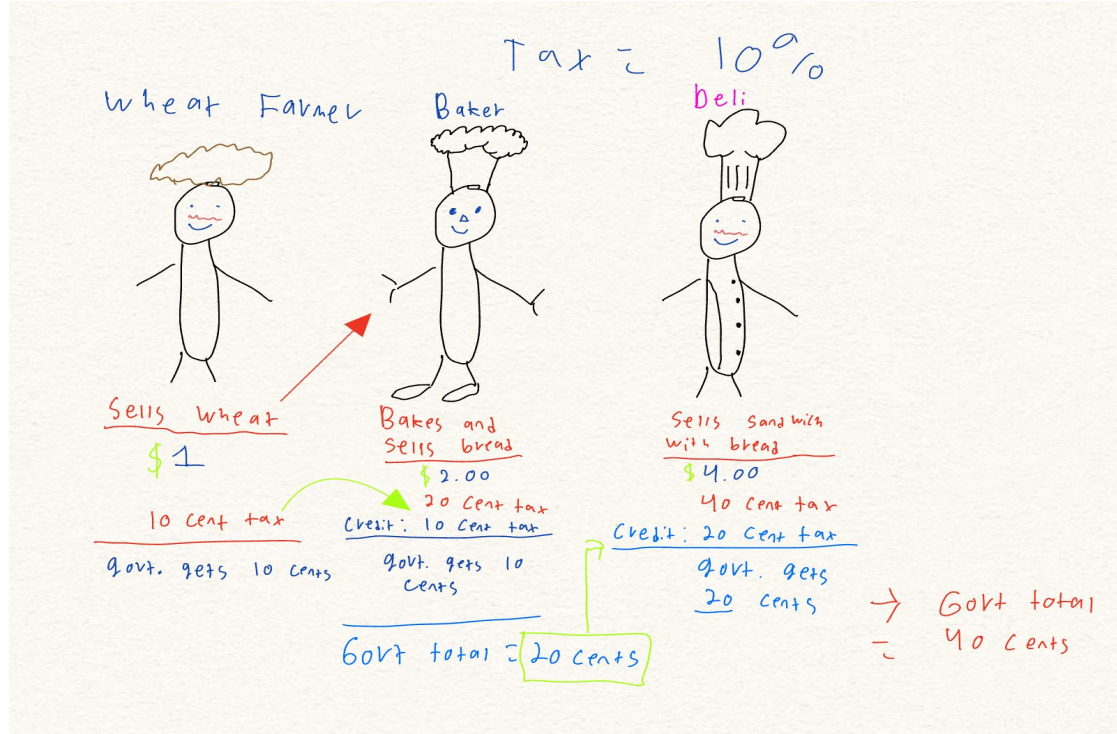
Tax AI services, not AI infrastructure

This avoids tax pyramiding: taxing value multiple times through the production chain.

Incentivizes productive capital investment.

Analogy: Tax the law firm's client invoice not the electricity or the computer the lawyer used.

HOW A VAT WORKS: THE SANDWICH EXAMPLE



KEY INSIGHT:

The government collected \$0.40 total.

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TAXING AI SERVICES: WHAT AND WHERE (Author suggestions)

API Usage

TAX ✓

Consumer uses AI assistant (API usage)

EXEMPT ✗

Business uses AI for internal ops

Robotics

TAX ✓

Household robot providing cleaning services to homeowners

EXEMPT ✗

Company buying robots for its fleet

Generative AI

TAX ✓

AI-generated text/images for end consumers

EXEMPT ✗

GPU ownership / server farm infrastructure

TAXING AI SERVICES: **CRITIQUE**

API Usage

EXEMPT ✗

Business use AI for internal ops

CRITIQUE

Companies incentive to move from L to K is maximized

Robotics

EXEMPT ✗

Company buying robots for its fleet

CRITIQUE

Again, L to K is maximized

Generative AI

EXEMPT ✗

GPU ownership / server farm infrastructure

CRITIQUE

If you exempt server farms from taxation, you're exempting the single largest capital investment AI companies make.

FINAL PHASE 1 THOUGHTS & TOOLS

FIXED-FACTOR TAXATION

Tax scarce resources that can't flee to tax havens: land, spectrum rights, orbital satellite slots, critical natural resources. Can generate 5–10% of GDP in revenue (estimation by Korinek).

DIFFERENTIAL COMMODITY TAXES

Ramsey (1927): higher taxes on goods harder to evade or produce at home. Restaurant meals > groceries. Yacht maintenance > basic transport. Entertainment > basic utilities.

CAPITAL TAXATION (LAST RESORT)

Keep as low as possible since it discourages investment, and only use if consumption taxes alone prove insufficient.

PHASE 2: THE OPTIMAL HARVESTING FRAMEWORK

"Just as a forest manager must decide how much timber to harvest each year to balance current needs against future growth, society must determine the optimal rate at which to tax AI capital."

THE KEY INSIGHT

Optimal tax rate on AI capital $\approx$ society's discount rate

Example: if society discounts future consumption at 4%, the optimal annual AI capital tax is 4%

Patient society (values future generations $>$) $\rightarrow$ tax lightly, let AI compound

Impatient society (values current needs $>$) $\rightarrow$ tax more, harvest value now

IMPLEMENTATION

Tax base metrics:

- Computing resources deployed
- Robotic capital deployed
- Total capital accumulation

Requires international coordination to prevent AI companies from relocating to tax havens

STRUCTURAL ALTERNATIVES TO TAXATION

The authors propose three non-tax mechanisms to capture AI's value for the public:

1

PUBLIC CO-INVESTMENT

Government provides data, infrastructure, or research support and takes equity stakes in return. Scales automatically with AI company success.

Ex: Norway's sovereign wealth fund from oil revenues

2

INFRASTRUCTURE-FOR-ACCESS DEALS

Instead of taxing, governments negotiate revenue-sharing agreements in exchange for providing valuable inputs: govt data, research facilities, test environments, regulatory clarity.

Ex.: spectrum auctions, mineral rights, toll road franchises.

3

AI DEVELOPMENT COMPACTS

Voluntary agreements where AI companies commit a % of gains to public benefit in exchange for regulatory clarity and infrastructure support.

CONNECTING TO GANS: WHAT THE TEXTBOOK TELLS US

POINTS TO BRING UP

Ch. 3: AI as a Prediction Machine:

AI lowers the cost of prediction. This generates lots of value but Gans shows the value is owned to whoever controls the AI, not the workers it replaced.

Ch. 4: Moving from Rules to Decisions:

When AI enables firms to make decisions instead of following rules, it reduces the need for judgment-based labor roles. This directly explains why labor income and therefore labor taxes declines.

A CRITIQUE

AI is imperfect prediction:

Gans Ch. 3 emphasizes that AI signals are never perfect they have false positive and false negative rates. Think of a doctor reading an X-ray vs. an AI reading it. AI might catch 90% of cases, but a 10% miss rate is catastrophic in medicine.

Korinek & Lockwood assume AI displacement is large and certain. But if AI prediction quality remains limited in all these complex domains (medicine, law, education) then humans stay very essential in these high stakes fields and the labor displacement will be slower than they forecast.

OUR CRITIQUE: DO CONSUMPTION TAXES PRESERVE WORK INCENTIVES?

THE AUTHORS' CLAIM:

"A consumption tax can replace an income tax that raises the same revenue without changing work incentives, since both reduce purchasing power by the same amount."

OUR PUSHBACK

The timing of when you feel the tax matters psychologically

Example scenario: You get a \$500 bonus at work. If your employer immediately takes \$150 before handing it to you, that stings in a way that motivates you to earn more next time. If you get the full \$500 but pay \$150 extra at the grocery store over the next month you barely notice it but it's the same total tax and a completely different psychological experience

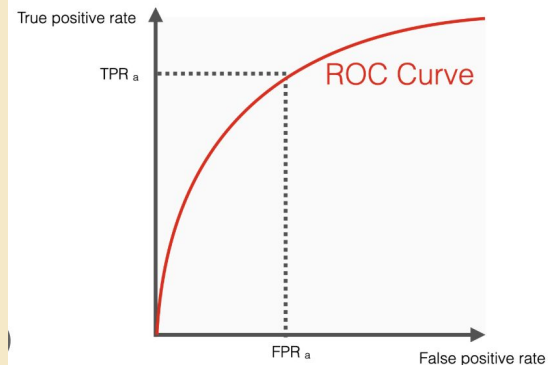
A tax you feel when you earn is more motivating to work harder than one you feel when you spend.

OUR CRITIQUE: AI IS IMPERFECT

GANS CH. 3: ROC CURVE FRAMEWORK

AI prediction is characterized by a tradeoff between true positive rates and false positive rates. Even the best AI models operate on this curve and they are never perfect. Think of a spam filter where if we crank it too high and you start blocking real emails. Dial it back and spam gets through. You're always on a tradeoff curve

EX:



THE FISCAL CRISIS

Korinek & Lockwood's Phase 1 and Phase 2 scenarios assume broad and rapid AI labor substitution. But if AI remains a complement rather than substitute for skilled professional judgment in many domains, the transition may be gradual enough to keep traditional tax systems in place

WHAT THE ESSAY GETS RIGHT

The structural problem is real

The tax rate gap between labor (~30%) and capital (~15-20%) is empirically documented. Labor's share of national income has already declined from 68% to 58% since 2000, and we can observe this happening now.

Act before the crisis, not during it

History supports proactive policy: nations that anticipated economic transitions (U.S. education investment, South Korea's knowledge economy pivot) worked better than those that reacted. Waiting for a fiscal emergency guarantees worse outcomes.

Tax consumption, not production inputs

The principle of taxing AI at the point of consumer delivery (not taxing GPUs, server farms, or B2B transactions) is smart. Think of a law firm that gets an AI to draft documents. It wouldn't be right to tax the AI as an equipment as that would be like taxing a computer at a trading desk. Instead the consumption tax is at the point where the drafts are made and handed to the clients which is the end user.

OUR VERDICT

Korinek & Lockwood identify a really interesting structural problem that many overlook when looking at AI capabilities

But two assumptions are pretty misleading:

1. **The behavioral assumption** that consumption taxes preserve work incentives equally to income taxes ignores how salience shapes motivation.
2. **The pace assumption** that AI will broadly displace professional labor may be too aggressive — Gans shows that in high-stakes domains, AI remains a complement to human judgment, not a substitute.

I think in this paper they are trying to be proactive in policy which is good but the specific design and assumptions of this AI boom for the future should also take more into the behavioral economics side and the limits of AI prediction

POINTS FOR CLASS DEBATE

GIVING AI COMPANIES ACCESS TO GOVERNMENT DATA?

Q1: Is government data actually the government's to give away?

THE MORALITY OF INFRASTRUCTURE-FOR-ACCESS DEALS

Q2: Should the government be able to regulate something they financially benefit from?